

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester MCA (CBCS) Degree Examinations
December 2024 – January 2025

22MCA331 – MOBILE COMPUTING AND APPLICATION DEVELOPMENT

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	Explain the key components of the GSM architecture.	5	L*1	1	2
	b)	What are the functions of the HLR and VLR in GSM?	5	L1	1	1,2
	c)	Discuss the evolution of wireless cellular networks from 1G to 5G, highlighting the key technologies and improvements.	10	L2	1	1
2.	a)	Describe the role of the BSS) in GSM architecture.	5	L2	1	2
	b)	Explain the key components of the GPRS network architecture.	5	L1	1	1
	c)	Explain the architecture of mobile computing, describing its key components and how they interact to provide mobile services.	10	L1	1	1,2
		Unit – II				
3.	a)	Explain the basic architecture of Android	5	L1	2	2
	b)	Explain the purpose of an Android Layout.	5	L1	2	2
	c)	Discuss the process of creating and applying custom views in Android. Provide a simple example to illustrate.	10	L2	2	1,3
4.	a)	What are the challenges in developing an Android application. Explain	5	L1	2	1
	b)	What is a Linear Layout in Android, and how does it arrange UI elements?	5	L1	2	2
	c)	Explain Android resource types and demonstrate how to apply resources in an application.	10	L2	2	1
		Unit – III				
5.	a)	What is the purpose of the activity lifecycle in Android?	5	L1	3	1
	b)	Differentiate between onPause() and onStop() in the Android activity lifecycle.	5	L2	3	2
	c)	What are Intents in Android? Explain the difference between explicit and implicit intents.	10	L1	3	2
6.	a)	What is the role of the onResume() method in the Android activity lifecycle?	5	L1	3	1
	b)	What is the significance of the onDestroy() method in the Android activity lifecycle?	5	L1	3	1
	c)	By providing an example explain how intents can be used in an application.	10	L3	3	1,2
		Unit – IV				
7.	a)	How would you implement a button click listener in Android to navigate to another activity?	5	L2	4	2
	b)	Describe how to implement a simple RecyclerView in Android.	5	L2	4	1
	c)	With an example, explain how you use a drawable resource in an Android XML layout file?	10	L3	4	1
8.	a)	How can you pass data from one activity to another in Android?	5	L2	4	1
	b)	What are the different types of user interaction in Android?	5	L1	4	1
	c)	How can you switch between different themes in an Android app at runtime?	10	L2	4	2
		Unit – V				
9.	a)	What is a Broadcast Receiver in Android, and how is it used?	5	L1	5	1
	b)	How do you start and stop a Service in Android?	5	L2	5	1

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c) How do you create and manage an SQLite database in Android? Provide an example.		10	L2	5	2
10.	a) What are the different types of Android Services, and what is their purpose?	5	L1	5	2
	b) What is a Notification in Android, and how is it structured?	5	L1	5	1
	c) How do you insert, update, delete, and query data using an SQLite database in Android?	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
